

Chapter 7 In-Class Exercise: Fourier Series as a Module

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1 Step 1: Create the Module

1. Previously, we created separate functions for one Fourier term, the Fourier sum, and the plot. We will now place those functions inside a module. Create a file called

`fourier_tools.py`

and begin with

```
import numpy as np
import matplotlib.pyplot as plt
```

Add the function that evaluates the n -th term:

$$a_n(x) = 2 \frac{(-1)^{n+1}}{n} \sin(nx).$$

```
def fourier_term(x, n):
    term = 2 * (-1)**(n + 1)
    term = term * np.sin(n*x) / n
    return term
```

2 Step 2: Add the Sum

1. Add a second function to the same module:

```
def sawtooth(x, N):
    total = 0.0
    for n in range(1, N + 1):
        total = total + \
            fourier_term(x, n)
    return total
```

Notice that `sawtooth()` calls another function located inside the same module.

- Which function performs one Fourier-series term?
- Which function controls how many terms are added?

3 Step 3: Add the Plot Function

1. Add a third function:

```
def plot_series(x, N):
    y = sawtooth(x, N)
    plt.plot(x, y, label=f"N = {N}")
```

The module now contains three related functions:

```
fourier_term()
sawtooth()
plot_series()
```

Save the completed file.

4 Step 4: Upload and Import

1. In Google Colab, upload `fourier_tools.py`. In a new code cell, import the module:

```
import fourier_tools
```

Also import the libraries needed by the main program:

```
import numpy as np
import matplotlib.pyplot as plt
```

Test one function directly:

```
print(fourier_tools.fourier_term(np.pi/4, 1))
```

5 Step 5: Main Program

1. Write the following script in a Colab code cell:

```
x = np.linspace(-np.pi, np.pi, 400)
fourier_tools.plot_series(x, 1)
fourier_tools.plot_series(x, 5)
fourier_tools.plot_series(x, 20)
plt.plot(x, x, label="f(x) = x")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.title("Sawtooth Fourier Series")
plt.legend()
plt.grid()
plt.show()
```

Run the program and inspect the graph.

- Which file contains the mathematical functions?
- Which part of the program determines N ?
- Change the final value from $N = 20$ to $N = 50$ and run the program again.

6 Challenge

Modify the main program so that the values of N are stored in `terms = [1, 5, 20, 50]` and use a `for` loop to call `fourier_tools.plot_series()` for each value.